

Customer Behavior Analysis

1. Project Overview

This project analyzes customer shopping behavior using transactional data from 3,900 purchases across various product categories. The objective is to identify spending patterns, customer segments, product preferences, and subscription behavior to support data-driven business decisions. The project integrates Python for data preprocessing and exploratory analysis, MySQL for structured business queries, and Power BI for interactive visualization.

2. Dataset Summary

Total Rows	3,900
Total Columns	18
Missing Values	37 values in Review Rating column

Key Features Overview:

Category	Features
Customer Demographics	Age, Gender, Location, Subscription Status
Purchase Details	Item Purchased, Category, Purchase Amount, Season, Size, Color
Shopping Behavior	Discount Applied, Promo Code Used, Previous Purchases, Frequency of Purchases, Review Rating, Shipping Type

3. Exploratory Data Analysis Using Python

Step	Description
Data Loading	Imported dataset using pandas library.
Initial Exploration	Used df.info() and df.describe() to inspect structure and statistics.
Missing Value Handling	Imputed missing Review Rating values using median rating per category.
Feature Engineering	Created age_group and purchase_frequency_days features.
Data Consistency & Integration	Removed redundant promo_code_used column and loaded cleaned data into MySQL.

4. Business Analysis Using SQL

Analysis Task	Purpose
Revenue by Gender	Compared total revenue generated by male and female customers.
High-Spending Discount Users	Identified discount users spending above average.
Top 5 Products by Rating	Determined highest average-rated products.

Shipping Type Comparison	Compared average purchase amounts between Standard and Express shipping.
Subscribers vs Non-Subscribers	Compared revenue and spending patterns.
Discount-Dependent Products	Identified products heavily reliant on discounts.
Customer Segmentation	Classified customers as New, Returning, or Loyal.
Top 3 Products per Category	Ranked most purchased items per category.
Repeat Buyers & Subscriptions	Analyzed subscription likelihood of repeat buyers.
Revenue by Age Group	Calculated revenue contribution by age segments.

5. Dashboard Development in Power BI

An interactive dashboard was developed in Power BI to visually represent revenue trends, customer segmentation, subscription impact, product performance, and shipping type comparisons. The dashboard enables dynamic filtering and drill-down capabilities for business stakeholders.

6. Conclusion

The Customer Behavior Analysis project successfully demonstrates how data analytics can drive informed business decisions. By integrating Python for data preprocessing, MySQL for structured querying, and Power BI for visualization, the project provides a comprehensive analytical workflow. The findings highlight the importance of subscription models, customer loyalty, optimized discount strategies, and demographic-based targeting. Overall, this analysis provides a scalable framework that can be extended with predictive modeling and advanced analytics for future business growth.